

Multi-RCM Kinematic Control for a ROSA-like Surgical Robot

A Unity Simulation of Double Remote-Center-of-Motion Control

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Abstract

This report presents a Unity-based simulation of a ROSA-like six-degree-of-freedom surgical robot implementing double Remote-Center-of-Motion (RCM) control, following the shaft-parameterization principle of Aghakhani et al. (ICRA 2013). A from-scratch damped-least-squares controller with null-space secondary objectives supports four operating modes and a three-phase safe-insertion sequence. Task 2 uses a strict tip-first hierarchy, a directional active-set constraint on joint 4, adaptive damping and bounded integration substeps. The controller is evaluated over five repeated nominal Unity simulations comprising 6850 logged samples and 423.21 s of simulation time. Using thresholds of 2 mm for entry-RCM stability, 3 mm for tip-target accuracy and 50 mm for cone tracking, all nine declared checks pass in every run, giving 45/45 successful run-level checks. The small between-run standard deviations demonstrate numerical repeatability under the tested nominal conditions, while the simulation-only model precludes hardware or clinical claims.

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1 Introduction and Clinical Motivation

Stereotactic neurosurgical procedures biopsy, electrode placement, catheter insertion require a rigid tool to reach a deep target through a single, pre-planned skull opening. Once the entry point is fixed, any lateral force exerted by the tool shaft against the edges of that opening risks tearing or bruising healthy tissue along the corridor. Robotic assistance can enforce this constraint far more precisely than a hand-guided or passively braced instrument: the robot’s controller can be made to guarantee that the tool axis always intersects the entry point, regardless of how the arm reconfigures to reach different targets.

The mechanism that formalizes this guarantee is the *Remote Center of Motion* (RCM): a virtual pivot point, external to the robot’s own mechanical structure, through which the tool shaft is constrained to pass at all times.

Unlike a mechanical RCM (a physical gimbal, as used in some laparoscopic robots), the RCM studied here is purely a *control* objective, implemented as an additional task in a task-priority kinematic controller.

This project implements and validates such a controller in Unity, for a ROSA-like six-degree-of-freedom manipulator, following the double-RCM task formulation of Aghakhani et al. (ICRA 2013), in which the RCM constraint can be located either at the entry point or, in an alternate mode, effectively relocated to the target itself.

2 The Remote-Center-of-Motion Concept

The RCM point is expressed as a point sliding along the tool shaft:

$$p_{\text{RCM}}(q, \lambda) = p_{\text{base}}(q) + \lambda L z(q), \quad (1)$$

where $p_{\text{base}}(q)$ is the position of the tool’s proximal reference point (a function of the joint configuration q), $z(q)$ is the tool’s axial unit vector, L is the tool length, and $\lambda \in [0.02, 0.995]$ is a dimensionless *penetration coordinate* that slides the candidate RCM point along the shaft as insertion proceeds.

The controller drives λ together with the joint velocities so that $p_{\text{RCM}}(q, \lambda)$ coincides with the desired entry (or target) marker, without requiring any physical hinge at that location.

Enforcing this constraint changes the character of the motion completely. Without it, nothing prevents the shaft from levering sideways against the entry point as the arm reconfigures to track a moving target. With the RCM task stacked alongside the tip task, the controller can move the tip to a new target while keeping the pivot locked at the entry, or in the alternate task modes studied here relocate the effective pivot to the target while the entry side sweeps a bounded cone. This is the “double RCM” idea referenced in the title: the same controller structure realizes an RCM at the entry *or* at the target, selected by which task mode is active.

3 System Description

3.1 Simulated Manipulator

The manipulator is a procedurally generated six-degree-of-freedom serial arm with a ROSA-like (plausible, not vendor-measured) Denavit–Hartenberg table, summarized in Table 1. Joint limits are symmetric except for the last joint (wrist roll), which is effectively unbounded ($\pm 360^\circ$).

Table 1: Denavit–Hartenberg parameters used for the procedural arm (as implemented in `ROSADoubleRCMController.cs`).

Link	a [m]	α [deg]	d [m]	θ_{offset} [deg]	Joint range [deg]
1	0.045	−90	0.42	0	[−170, 170]
2	0.42	0	0.00	0	[−120, 120]
3	0.055	−90	0.00	0	[−150, 150]
4	0.000	90	0.36	0	[−170, 170]
5	0.000	−90	0.00	0	[−120, 120]
6	0.000	0	0.12	0	[−360, 360]

The scene, arm, needle tool, and target/entry markers are all built procedurally at runtime by a single script (`Project4SceneBuilder.cs`), rather than hand-assembled in the Unity editor, keeping the project self-contained and reproducible. A second script (`FreeFlyCameraKeyboard.cs`) provides an independent keyboard fly-camera for visual inspection and screen recording; it has no effect on the controller or the logged data.

Figure 1 shows the complete Unity simulation environment. The procedurally generated ROSA-like manipulator controls the needle-like tool relative to the entry and target markers. The translucent sphere represents the simplified skull geometry, while the runtime overlay reports the active task, insertion phase, instantaneous tracking errors, joint-limit status, and available keyboard commands.

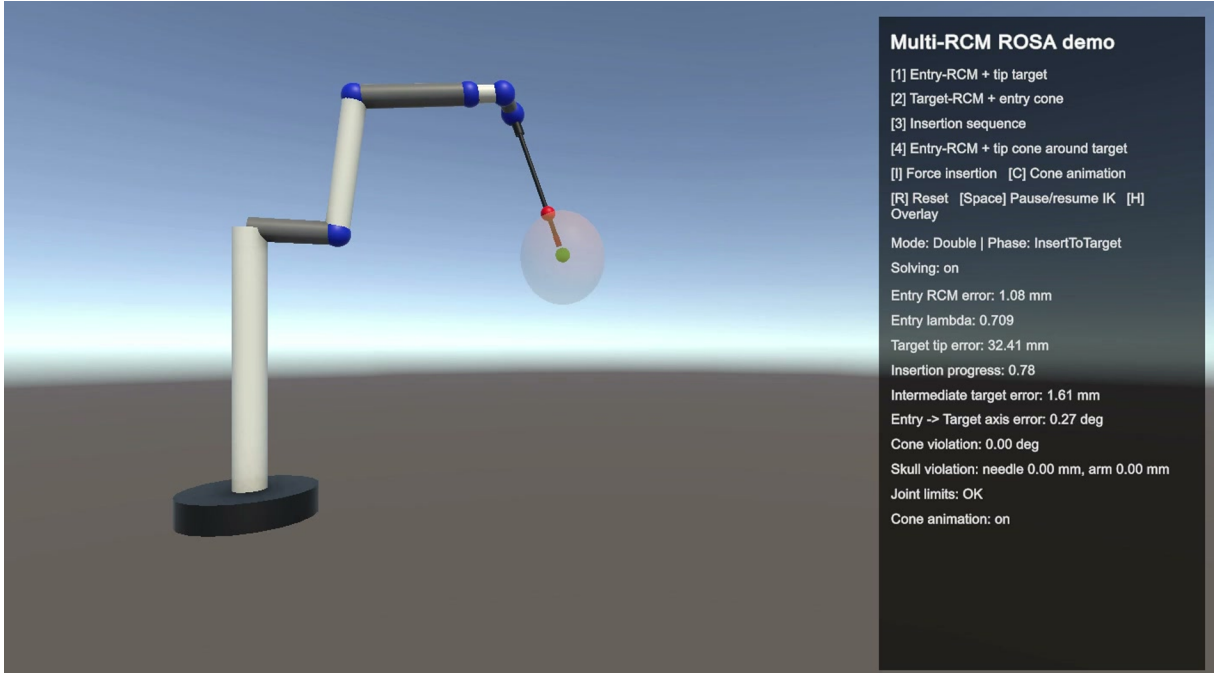


Figure 1: Unity simulation overview during the `InsertToTarget` phase of Task 3. The screenshot shows the ROSA-like manipulator, the needle, the entry and target markers, the simplified skull geometry, and the runtime diagnostic overlay. The displayed values are instantaneous transient values; the quantitative results reported in Section 7 are obtained from the five complete CSV logs.

3.2 Task Modes

All logic kinematics, the control law, the insertion state machine, the on-screen overlay, and CSV logging lives in a single controller script, `ROSADoubleRCMController.cs`. It exposes five

mutually exclusive modes, four of which are active task modes (Table 2); the fifth, **Hold**, freezes the arm and is not part of the validated task set.

Table 2: Task modes implemented in the controller.

Task	Enum name	Description
T1	<code>EntryRCM_TipTarget</code>	RCM locked at the entry point while the tip tracks a fixed target.
T2	<code>TargetRCM_EntryCone</code>	RCM relocated to the target while the entry side sweeps a bounded cone.
T3	<code>InsertionSequence</code>	Three-phase safe-insertion sequence (the main clinical task, Section 3.3).
T4	<code>EntryRCM_TipConeAroundTarget</code>	RCM locked at the entry while the tip sweeps a bounded cone around the target.

3.3 Task 3: The Safe-Insertion Sequence

Task 3 is a three-phase finite-state machine that mirrors the intended clinical workflow (a fourth internal state, **Done**, marks completion):

1. **ApproachEntry**: the needle axis is aligned with the entry–target line while the tip approaches from outside the trocar; the RCM constraint is *not yet* active.
2. **PierceEntry**: the tip travels from the outside approach point to the entry (trocar) point.
3. **InsertToTarget**: the RCM is locked at the entry point and the tip advances along the shaft axis to the target, terminating in the **Done** state once both the insertion progress and the tip-to-target distance fall below fixed thresholds.

4 Kinematic Control Framework

4.1 Forward Kinematics and Numerical Jacobians

For the six revolute joints $q \in \mathbb{R}^6$, forward kinematics is obtained from the standard product of the Denavit–Hartenberg transforms,

$$T_0^6(q) = \prod_{i=1}^6 A_i(q_i). \quad (2)$$

It provides the proximal tool point $p_{\text{base}}(q)$ and the unit tool axis $z(q)$. A generic point on the rigid tool is therefore

$$p_{\text{tool}}(q, s) = p_{\text{base}}(q) + sLz(q), \quad s \in [0, 1], \quad (3)$$

with the physical tip at $s = 1$. The implementation uses one generic numerical pipeline for every controlled point: tip, RCM point, cone reference point and the guide point used during insertion. Their Jacobians are computed using central finite differences on the seven-component extended state $x = (q, \lambda) \in \mathbb{R}^7$. This avoids maintaining a separate analytic Jacobian for each phase-dependent task and permits the DH table or task-point definitions to be changed without reformulating the controller.

4.2 RCM Extended State

The candidate remote center of motion slides along the fixed-length tool segment according to

$$p_{\text{RCM}}(q, \lambda) = p_{\text{base}}(q) + \lambda L z(q), \quad \lambda \in [0.02, 0.995]. \quad (4)$$

The penetration coordinate λ is solved and integrated together with the six joint coordinates. Selecting the desired RCM reference as either the entry point or the target permits the same geometric model to implement the entry-centred and target-centred modes. In this project the RCM condition is a gain-weighted Cartesian regulation task. It is therefore enforced to high accuracy after convergence, but it is not a mathematically hard equality constraint at every transient sample.

4.3 Task-Space Regulation and Damped Least Squares

For every active point task $y_i = f_i(q, \lambda)$, the Cartesian error is converted into a magnitude-limited proportional velocity command,

$$e_i = y_{i,d} - y_i, \quad v_i = K_i \text{sat}_{\varepsilon_i}(e_i). \quad (5)$$

Except for the strict-priority formulation used in Task 2, the active Jacobians and velocity commands are stacked as $J = [J_1^T \cdots J_m^T]^T$ and $v = [v_1^T \cdots v_m^T]^T$. The extended-state velocity is obtained through damped least squares [2]:

$$\dot{x}_{\text{base}} = J^T (J J^T + \mu^2 I)^{-1} v, \quad x = (q, \lambda), \quad \mu = 0.055. \quad (6)$$

The solver, including the linear-system solution, is implemented directly in C# without an external inverse-kinematics package. The base gains are $K_{\text{tip}} = 3.5$, $K_{\text{RCM}} = 17.0$ and $K_{\text{cone}} = 6.0$, with mode-specific multipliers used to reflect the relative importance of the active objectives.

A secondary command w combines joint-limit avoidance, regulation of λ toward its mode-dependent nominal value, and the available skull-avoidance bias. It is projected through a damped approximation of the null space,

$$\dot{x} = \dot{x}_{\text{base}} + s_{\text{ns}} (I - J_{\mu_b}^+ J) w. \quad (7)$$

This term improves the internal configuration without intentionally changing the commanded Cartesian task. The skull-related component is conditional on the corresponding simulation option; consequently, the logged geometric skull/corridor check must be interpreted as a simulation-level diagnostic and not as a force- or anatomy-based safety guarantee.

4.4 Priority, Joint Constraints and Numerical Integration

Tasks 1, 3 and 4 use gain-weighted stacking. Task 2 instead applies strict task priority [3]: the physical tip is stabilized first, a damped tip null-space projector is constructed, and the entry-cone command is solved inside that projected space. A final correction reduces any residual effect of the secondary motion on the primary objective.

Task 2 also contains a directional active-set guard for joint 4. When this joint enters a 25° soft margin and the computed velocity points farther toward the physical limit, its Jacobian column is removed, $\dot{q}_4$ is set to zero and the Cartesian command is redistributed over the remaining joints and λ . Motion away from the limit remains permitted, and the joint 4 null-space contribution is suppressed while the guard is active. This preserves the primary tip constraint without permanently locking the wrist.

Near singular configurations are handled without an explicit SVD. If the unsaturated solution would exceed the per-joint speed bound, the solve is repeated with increased damping before hard velocity saturation. The final limits are $|\dot{q}_i| \leq 80^\circ/\text{s}$ and $|\dot{\lambda}| \leq 0.55 \text{ s}^{-1}$. Explicit-Euler integration is then performed using substeps no longer than 8 ms:

$$q_{k+1} = \text{clamp}(q_k + \dot{q} \Delta t), \quad \lambda_{k+1} = \text{clamp}(\lambda_k + \dot{\lambda} \Delta t, 0.02, 0.995). \quad (8)$$

5 Implemented Surgical Tasks

Table 3: Controlled point tasks in the four implemented operating modes.

Task	Controlled objectives	Solution structure
T1	Tip to fixed target; RCM to entry point	Gain-weighted stack
T2	Tip fixed at target; entry-side point to cone	Strict tip-first priority
T3	Phase-dependent approach, piercing and insertion objectives	Gain-weighted stack
T4	RCM to entry point; tip to cone around target	Gain-weighted stack

5.1 Task 1: Entry-Centred RCM with Fixed Tip Target

Task 1 represents the conventional entry-port configuration. The RCM point is regulated at the surgical entry while the physical tool tip is driven to the deep target. Both objectives are active simultaneously. The higher base RCM gain emphasizes preservation of the entry pivot while still permitting the tip to converge to the target.

5.2 Task 2: Target-Centred Mode with Entry-Side Cone

Task 2 keeps the physical tip fixed at the target as the primary objective and moves an entry-side point along a bounded conical reference as the secondary objective. The cone is defined in a plane orthogonal to the entry–target axis,

$$p_{\text{cone},2}(t) = p_{\text{entry}} + r_2 (\cos \phi b_1 + \sin \phi b_2), \quad \phi(t) = \omega_2 t, \quad (9)$$

where b_1 and b_2 span the transverse plane. In the implemented setup $r_2 \simeq 8.79$ mm and $\omega_2 = 8^\circ/\text{s}$. The tip command is solved first and the cone motion is projected into its damped null space. The logged `target_rcm_error_mm` is therefore an auxiliary geometric diagnostic, not one of the declared actively minimized validation quantities; the validated primary quantity is the physical tip-to-target error.

5.3 Task 3: Three-Phase Safe Insertion

Task 3 implements the clinical workflow as a finite-state sequence:

1. **ApproachEntry**: the tool axis is aligned with the entry–target direction and the tip approaches a pre-entry point outside the skull model;
2. **PierceEntry**: the tip progresses from the external approach point to the entry marker, with RCM pre-stabilization introduced near the end of the phase;
3. **InsertToTarget**: the RCM is regulated at the entry while the tip advances toward the deep target; completion leads to the internal **Done** state.

A continuously active guide-point objective promotes collinearity between the shaft and the entry–target line throughout the active phases. This avoids a separate alignment-only state and reduces unnecessary state transitions.

5.4 Task 4: Entry-Centred RCM with Tip Cone

Task 4 retains the conventional RCM at the entry point while the tip follows a cone around the deep target,

$$p_{\text{cone},4}(t) = p_{\text{target}} + r_4 (\cos \phi b_1 + \sin \phi b_2). \quad (10)$$

The implemented radius is $r_4 \simeq 13.8$ mm and the angular speed is $\omega_4 = 7^\circ/\text{s}$. Mode-specific gain and null-space multipliers favor entry-pivot stability over perfect tracking of the moving secondary reference. Transitions between Tasks 1–4 are operator-triggered; only the internal Task 3 phase transitions are autonomous. Cone motion uses a short warm-up interval after activation so that the controller can first settle the fixed-point objective.

6 Experimental Methodology

6.1 Data Source

The quantitative evaluation is based on five repeated Unity simulations acquired under the same controller configuration and task sequence.

Together, the logs contain 6850 samples and 423.21 s of recorded simulation time.

Individual run durations range from 80.56 to 86.60 s. Each run executes the modes in the order $T3 \rightarrow T1 \rightarrow T2 \rightarrow T4 \rightarrow \text{Hold}$. All quantitative results presented in this section are derived from these five logs.

The five runs are treated as five repeated experiments. Individual time samples within one log are not treated as independent repetitions. This distinction avoids artificially inflating the sample size and makes the reported between-run variation directly interpretable as numerical repeatability under the tested conditions.

6.2 Data Processing and Integrity Verification

The analyzer first verifies and merges the CSV structure while retaining the source filename as a run identifier. It then evaluates every check independently for each run and produces a conservative pooled result using the worst settled value observed across all runs. The generated per-run checks, repeatability statistics, phase summaries, and combined data are stored with the analysis outputs.

Task-specific plots use local task time, with the start of each task aligned at $t = 0$ for every run. This representation allows direct comparison of convergence profiles despite small differences in manually selected task durations. A single concatenated multi-run timeline is not used because the five logs have independent time origins.

Similarly, the global “active task error” is not used for validation because its physical meaning and applicable threshold change when the active task changes.

6.3 Validation and Repeatability Metrics

Nine pass/fail checks are evaluated with a 1 s settling exclusion after each relevant task or phase transition. The thresholds are entry-RCM stability ≤ 2 mm, tip-target tracking ≤ 3 mm, entry-/tip-cone tracking ≤ 50 mm, and skull/corridor violation ≤ 0 mm.

For each check, the analysis reports the settled maximum for every run, the mean and sample standard deviation of the five run maxima, the best-to-worst range, and the run-level pass rate. Raw maxima are retained in the time-series figures and may be higher than the settled maxima used by the automatic checks. This is expected when a new task or phase is activated while the robot is still converging from the preceding configuration.

Because the five simulations use the same nominal setup and initial conditions, close agreement between their trajectories demonstrates numerical repeatability. It does not by itself establish robustness to changes in anatomy, target position, initialization, model parameters, measurement noise, or disturbances.

7 Results

7.1 Overall Validation and Repeatability

The controller was evaluated over five repeated simulations performed under the same nominal conditions. For every run, each validation metric was evaluated using its maximum value after a 1 s settling exclusion following the relevant task or phase transition. This prevents the automatic pass/fail decision from being dominated by the short activation transients that are nevertheless shown in the time-history plots.

Table 4 summarizes the nine declared validation checks. The mean and sample standard deviation are computed across the five per-run settled maxima, whereas the worst-run value provides the conservative comparison with the acceptance threshold. All 45 run-level checks pass.

Table 4: Validation and repeatability results over five simulation runs. SD is the sample standard deviation of the five settled maxima.

Check	Threshold [mm]	Mean maximum $\pm$ SD [mm]	Worst run [mm]	Passed runs	Result
T1 entry-RCM stability	2	0.101 ± 0.014	0.122	5/5	PASS
T1 tip reaches target	3	0.462 ± 0.066	0.560	5/5	PASS
T2 entry-cone tracking	50	5.981 ± 0.051	6.018	5/5	PASS
T2 tip fixed at target	3	0.008 ± 0.001	0.008	5/5	PASS
T3 insertion entry-RCM	2	0.239 ± 0.000	0.239	5/5	PASS
T3 final target error	3	1.537 ± 0.019	1.561	5/5	PASS
T3 skull/corridor safety	0	0.000 ± 0.000	0.000	5/5	PASS
T4 entry-RCM stability	2	0.282 ± 0.001	0.283	5/5	PASS
T4 tip-cone tracking	50	2.754 ± 0.010	2.763	5/5	PASS

Figure 2 reports the worst settled result for each positive-threshold check, normalized by its own threshold. The dashed line at 100% is the acceptance boundary. Every bar remains well below this boundary. The most demanding check is the Task 3 final target error, whose worst result is 1.561 mm against 3 mm, or approximately 52% of its allowed value. The zero-threshold skull/corridor check cannot be normalized by division and is therefore reported separately in Table 4.

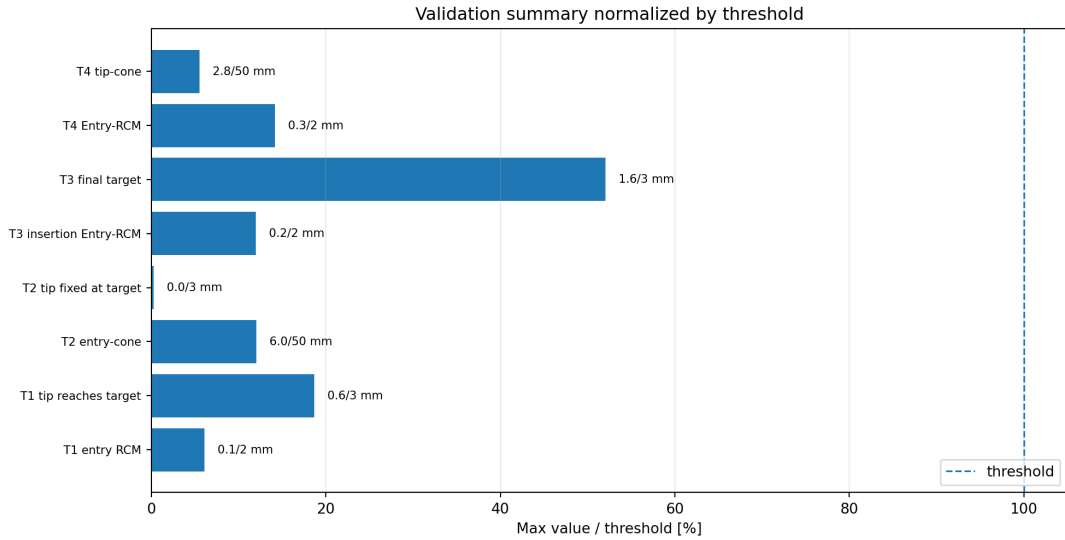


Figure 2: Conservative validation summary. Each bar is the worst settled value across the five runs, normalized by the corresponding positive threshold.

Figure 3 shows the settled maximum of every individual run. Colored markers identify the

five runs, the black diamond is their mean, and the horizontal black bar is the sample standard deviation. The tight clusters demonstrate highly repeatable numerical behavior for the tested nominal setup. This result establishes repeatability under repeated nominal conditions, rather than robustness to changes in anatomy, initial conditions, measurement noise, or target location.

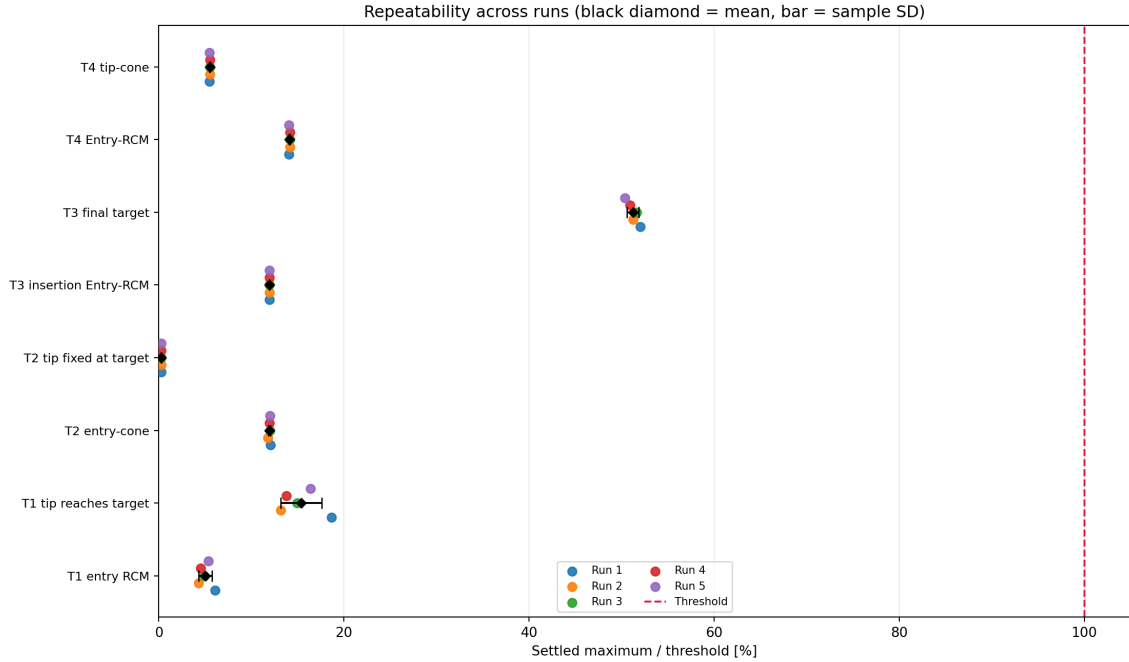


Figure 3: Repeatability of the positive-threshold validation checks. Colored markers denote individual runs; the black diamond and horizontal bar denote the mean and sample standard deviation.

7.2 Task 1: Entry-RCM with Tip Target

Task 1 simultaneously regulates the remote center of motion at the entry point and the physical tool tip at the deep target. Figure 4 shows that the entry-RCM error remains approximately one order of magnitude below the 2 mm threshold. The mean settled maximum is 0.101 ± 0.014 mm and the worst run reaches 0.122 mm. The small separation between the five curves is also reflected by their low absolute dispersion.

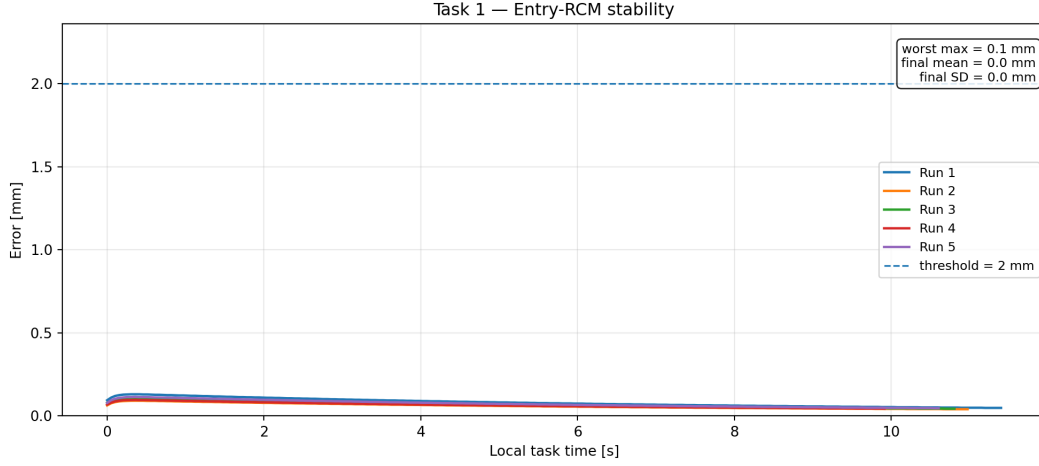


Figure 4: Task 1 entry-RCM error. The worst settled maximum is 0.122 mm against the 2 mm threshold.

The tip-target error in Figure 5 decreases throughout the task. Its mean settled maximum is 0.462 ± 0.066 mm and its conservative worst value is 0.560 mm, remaining below 19% of the 3 mm threshold. The final values are approximately 0.2 mm. Task 1 therefore achieves both entry pivot stabilization and accurate target positioning in every run.

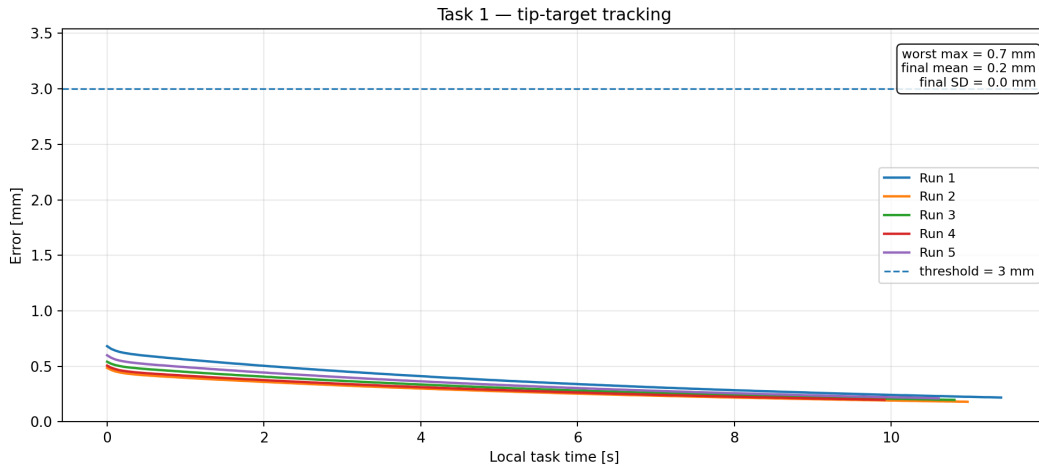


Figure 5: Task 1 tip-target tracking. The worst settled maximum is 0.560 mm against the 3 mm threshold.

7.3 Task 2: Tip-Fixed Target with Entry-Side Cone

Task 2 assigns strict priority to maintaining the physical tool tip at the target, while an entry-side point follows a rotating conical reference as a secondary null-space objective. Figure 6 shows that the entry-cone error is bounded throughout the task. Its raw worst value is approximately 8.8 mm, already well below the 50 mm limit. After settling, the mean of the five maxima is 5.981 ± 0.051 mm and the worst run reaches 6.018 mm, corresponding to only 12% of the available tolerance. The final mean is approximately 4.0 mm; the small final dispersion of approximately 0.4 mm is consistent with repeatable tracking of the moving reference.

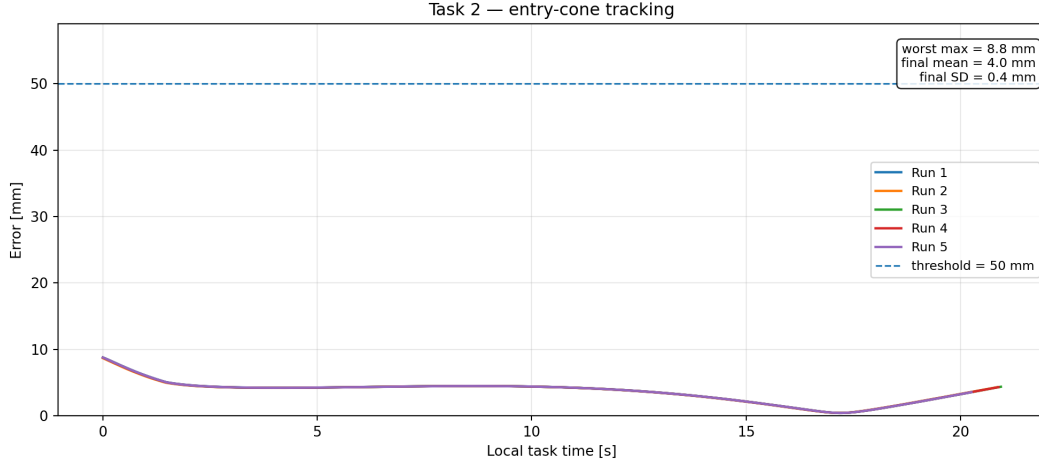


Figure 6: Task 2 entry-cone tracking. The raw worst value is approximately 8.8 mm and the worst settled maximum is 6.018 mm against the 50 mm threshold.

The primary tip-target objective is considerably more accurate. As shown in Figure 7, a short activation transient of approximately 0.2 mm rapidly vanishes. The mean settled maximum is 0.008 mm and every run has a maximum between 0.007 and 0.008 mm, compared with the 3 mm threshold. The result confirms that the hierarchical formulation preserves the surgical tip position while the secondary entry-side motion is executed.

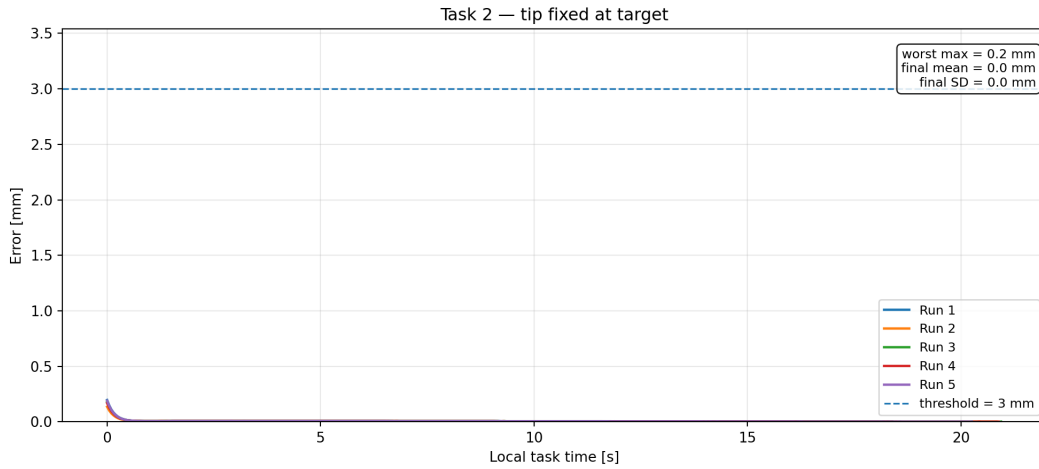


Figure 7: Task 2 primary tip-target error. The worst settled maximum is 0.008 mm against the 3 mm threshold.

7.4 Task 3: Safe Insertion

Task 3 performs the complete approach and insertion procedure. The robot must align the tool with the planned entry-target direction, pass through the entry point, preserve the entry RCM during insertion, reach the deep target, and remain inside the permitted skull corridor.

Figure 8 shows a raw Entry-RCM error of approximately 5.4 mm at phase activation. This is a short transition transient: the error falls below the 2 mm threshold in less than one second and subsequently remains close to zero. The settled maximum is 0.239 mm in every run, corresponding to approximately 12% of the allowed value. The essentially coincident curves indicate deterministic and repeatable entry-pivot regulation.

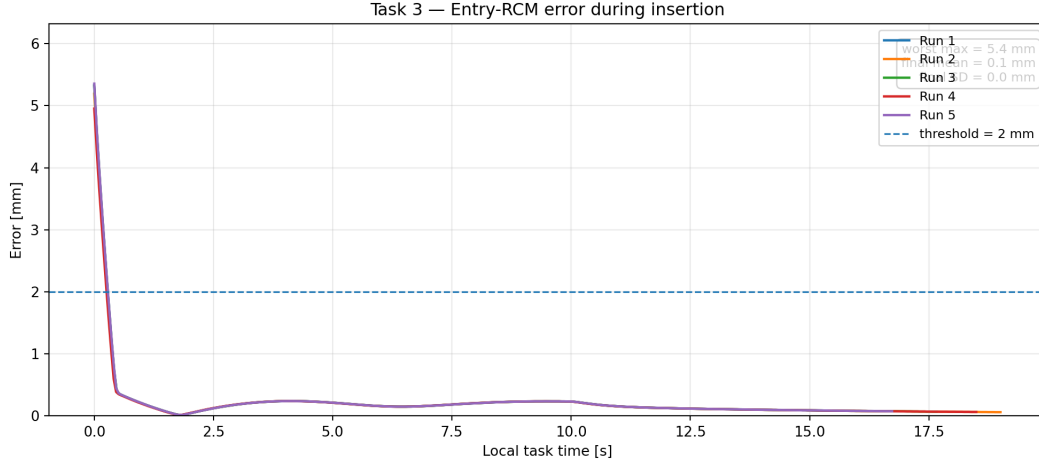


Figure 8: Task 3 Entry-RCM error during insertion. The raw activation transient is approximately 5.4 mm; the worst settled maximum is 0.239 mm against the 2 mm threshold.

Figure 9 reports the tip-target error during the completed insertion phase. The initial value of approximately 5.8 mm falls below the 3 mm threshold during the initial convergence interval and continues to decrease thereafter. The mean settled maximum is 1.537 ± 0.019 mm and the worst run reaches 1.561 mm. The final mean is approximately 0.6 mm with about 0.1 mm standard deviation. This is the closest of the declared checks to its threshold, but it still retains a margin of approximately 1.44 mm in the worst run.

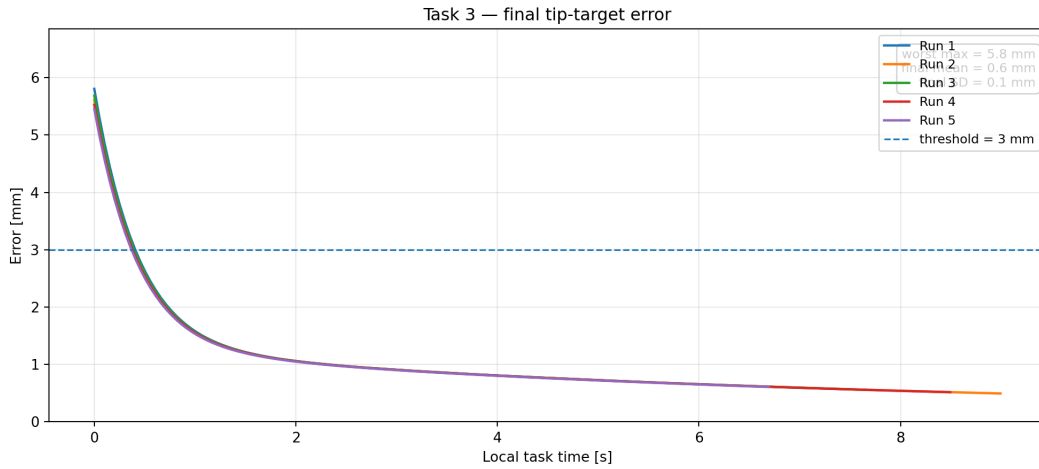


Figure 9: Task 3 final tip-target error. The worst settled maximum is 1.561 mm against the 3 mm threshold, and the final mean is approximately 0.6 mm.

No skull/corridor violation is recorded in any sample of any run, as shown in Figure 10. The 10^{-6} scale displayed automatically by the plotting library does not indicate a small nonzero error: all plotted values are exactly 0.000 mm.

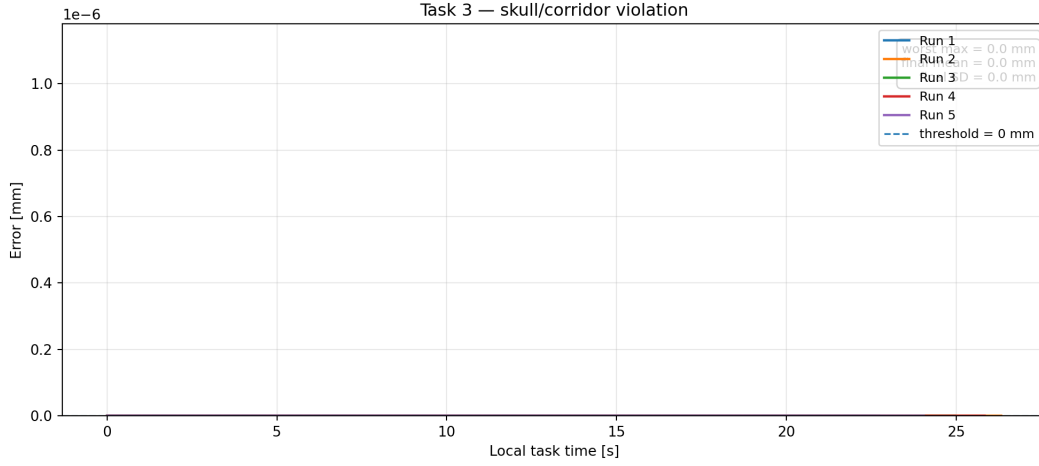


Figure 10: Task 3 skull/corridor violation. The recorded value is exactly 0.000 mm throughout all five runs.

The tip-to-line distance in Figure 11 evaluates alignment with the finite entry-target segment. The initial value of approximately 151.6 mm belongs to the external approach, when the tip has not yet reached the planned surgical line, and is not interpreted as an intracranial corridor violation. The distance converges to approximately zero as the insertion trajectory is acquired and remains close to the planned line for the rest of the sequence.

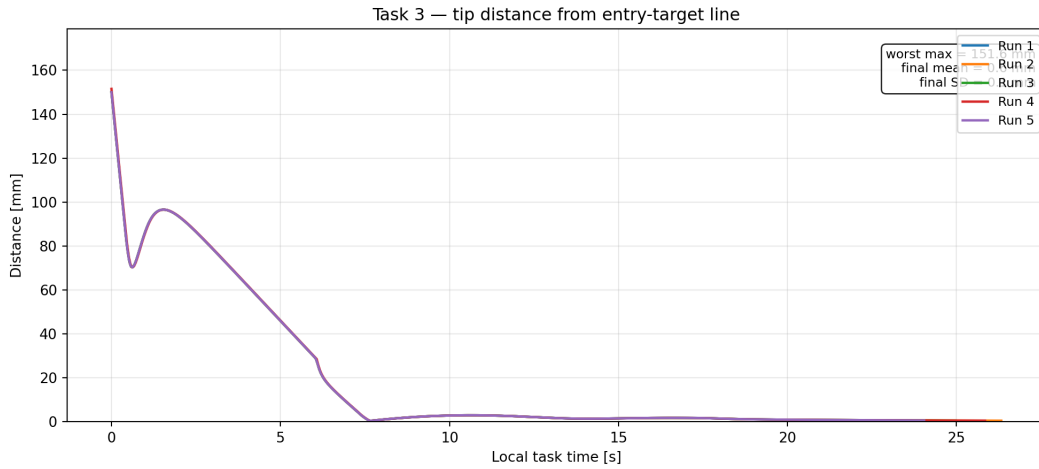


Figure 11: Task 3 distance between the tool tip and the finite entry-target segment. The large initial value belongs to the external approach; the final distance is approximately zero.

Figure 12 provides the corresponding orientation measure. The tool-axis error initially reaches approximately 33.3° and then decreases continuously as the tool aligns with the entry-target direction. It approaches zero at the end of Task 3 in all five runs. The positional and angular diagnostics therefore provide mutually consistent evidence of successful trajectory acquisition.

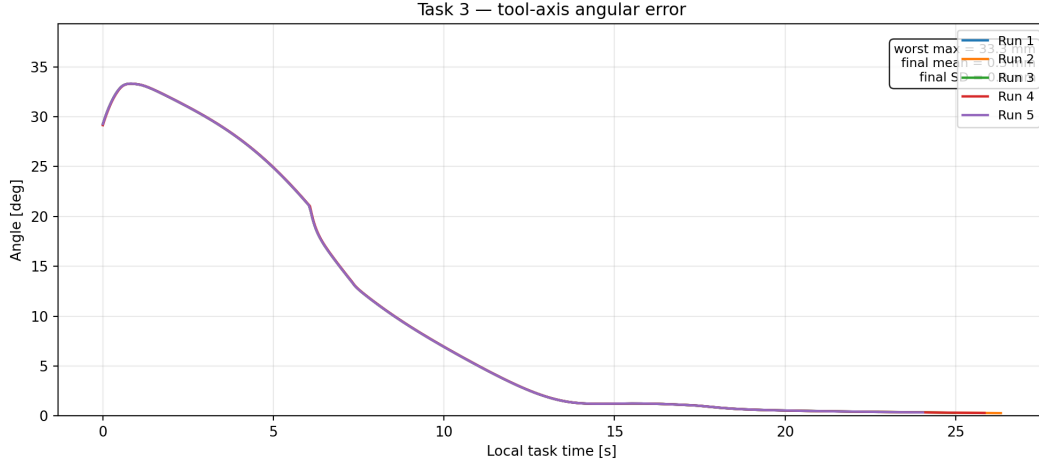


Figure 12: Task 3 angular error between the tool axis and the planned entry-target direction. The error converges from approximately 33.3° to approximately zero.

7.5 Task 4: Entry-RCM with Tip Cone

Task 4 keeps the entry RCM fixed as its primary objective while the tool tip tracks a circular cone around the deep target as a secondary objective. Figure 13 contains a raw task-activation transient of approximately 9.4 mm. The error rapidly falls below 2 mm and remains small after settling. The mean settled maximum is 0.282 ± 0.001 mm and the worst run reaches 0.283 mm, or approximately 14% of the entry-RCM threshold. The final error is approximately 0.1 mm.

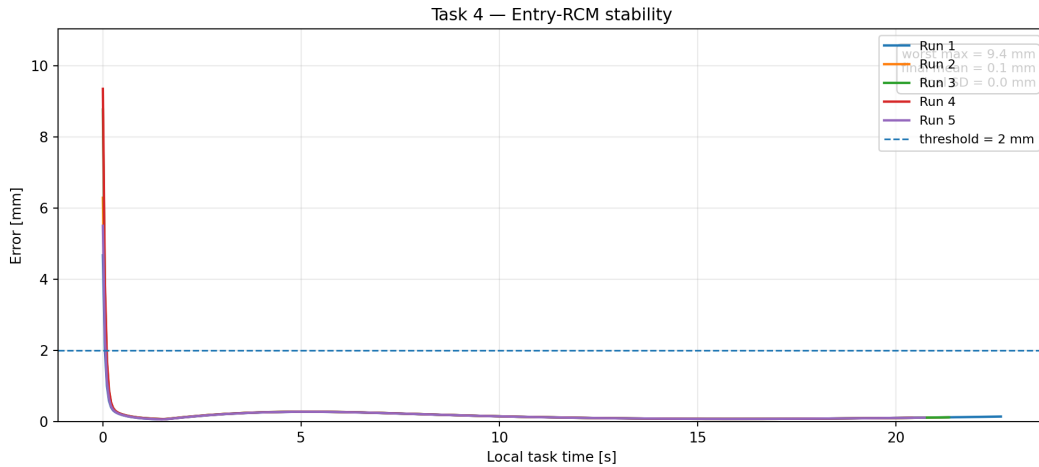


Figure 13: Task 4 entry-RCM stability. The raw activation transient is approximately 9.4 mm; the worst settled maximum is 0.283 mm against the 2 mm threshold.

The tip-cone tracking error in Figure 14 has a raw worst value of approximately 14.1 mm, which is already below the 50 mm limit. The mean settled maximum is 2.754 ± 0.010 mm and the worst run reaches 2.763 mm, using only about 5.5% of the allowed tolerance. The final mean is approximately 1.5 mm with a standard deviation of about 0.1 mm. Task 4 therefore preserves the stricter entry-pivot objective while accurately executing the secondary moving-tip trajectory.

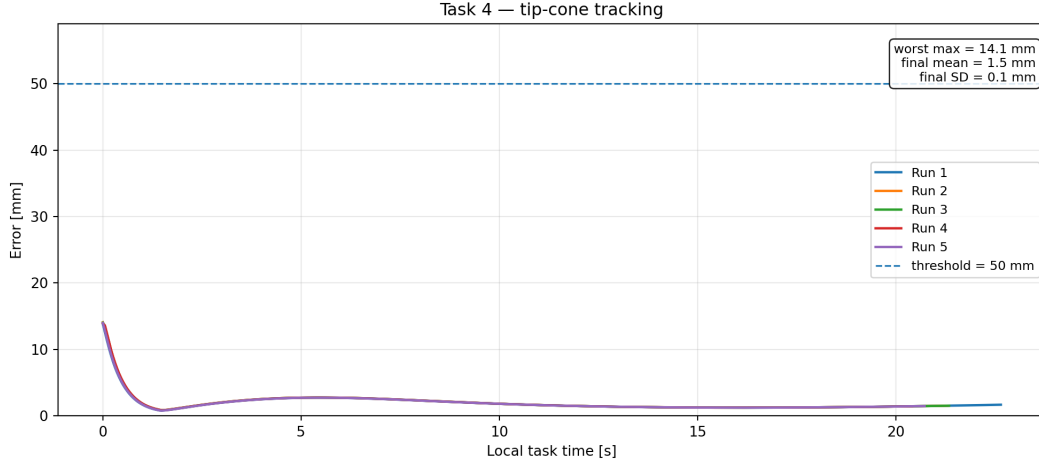


Figure 14: Task 4 tip-cone tracking. The raw worst value is approximately 14.1 mm and the worst settled maximum is 2.763 mm against the 50 mm threshold.

7.6 Results Summary

All nine declared validation checks pass in all five runs, yielding 45 successful run-level checks. Task 1 provides sub-millimetric entry-RCM and tip-target regulation. Task 2 maintains the tip essentially fixed at the target while tracking the entry-side cone within a wide margin. Task 3 converges to the planned insertion line, maintains the entry RCM, reaches the final target, and records no skull/corridor violation. Task 4 preserves the entry pivot while tracking the moving tip cone. The small sample standard deviations demonstrate repeatable numerical behavior for repeated nominal simulations.

8 Comparison with the ICRA 2013 RCM Approach

Table 5 compares the implemented controller with the RCM formulation presented by Aghakhani et al. [1]. The two works share the central geometric idea of augmenting the robot state with a shaft coordinate λ , but they differ in application, constraint formulation and implementation choices. The comparison is qualitative: the different robots, tasks and simulation environments prevent a meaningful comparison of absolute numerical accuracy.

Table 5: Comparison between the ICRA 2013 RCM formulation and this project.

Aspect	Aghakhani et al., ICRA 2013	This project
Application	Endoscopic-camera visual servoing and constrained path following	Needle positioning and insertion for a neurosurgical scenario
Robot platform	KUKA KR5 in Webots and DLR MIRO examples	Procedural ROSA-like 6R arm in Unity
RCM geometry	Point sliding along a robot-link segment through coordinate λ	Point sliding along the rigid tool shaft through coordinate λ
RCM placement	Fixed entry/trocar RCM	Entry-centred modes (T1, T3, T4) and target-centred extension (T2)
Constraint structure	RCM included as a hard velocity constraint in an extended Jacobian	Gain-weighted RCM regulation; strict hierarchy used for the T2 tip objective
Jacobians	Analytic formulation	Central finite-difference Jacobians for all task points
Singularity handling	Not the main focus of the reported formulation	DLS with adaptive damping based on the unsaturated joint-speed demand
Joint limits	Not explicitly developed in the reported examples	Hard clamping, null-space avoidance and directional Task-2 joint-4 guard
Safety model	RCM constraint and obstacle-avoidance example	Geometric skull/corridor diagnostics in a simplified Unity anatomy
Validation	Simulation demonstrations	Five repeated nominal simulations; 45/45 declared run-level checks passed
Main limitation	No hardware or clinical validation	Kinematic-only, simplified geometry and no hardware or clinical validation

The conventional trocar constraint in the reference paper corresponds most directly to the entry-centred modes T1, T3 and T4. Task 2 is an extension of the same shaft-parameterization principle: the physical tip is held at the target while the secondary motion is generated at the entry side. Switching between these placements requires no new geometric model because the controller only changes the active task points and their priority.

The reference method gives the RCM equality a structurally hard role inside the extended Jacobian. The present implementation treats it as a high-gain point-regulation task in Tasks 1, 3 and 4, which simplifies live switching between four modes but permits short activation transients. The five-run results show that these transients decay rapidly and that every settled RCM check remains comfortably below its threshold. This empirical result does not turn the soft task into a formal hard constraint; it only validates the chosen gains for the tested nominal configuration.

8.1 Rationale for the Implementation Choices

Generic numerical Jacobians. The controller contains several phase-dependent task points.

A central finite-difference routine automatically covers each point and remains valid when the DH table or task definitions change. This improves maintainability at the cost of additional forward-kinematics evaluations and some numerical sensitivity.

Strict priority only where required. Task 2 has a clearly asymmetric objective: the tip must remain fixed while the entry-side point moves. A tip-first hierarchy is therefore used to prevent the secondary cone task from degrading the surgical target. In the other modes, the active objectives must converge together and are managed through tuned gain weighting.

Adaptive DLS rather than an SVD-based index. The ratio between requested and allowable joint speed provides an inexpensive indication that the current solve is becoming poorly conditioned. Increasing damping before saturation reduces aggressive commands without adding a singular-value decomposition to every integration substep.

Directional active-set handling. The joint 4 guard constrains only outward motion near the limit and still permits recovery motion. This is less restrictive than permanently removing the joint and directly protects Task 2 from wrist saturation while retaining the remaining redundancy.

Mode-specific gains and cone speeds. The moving cone references compete with fixed-point accuracy. Their gains and angular speeds were selected so that the fixed RCM or tip objective remains dominant. The resulting performance is documented by the five-run analysis, but the tuning is scenario-specific and is not claimed to be globally optimal.

9 Limitations

- The manipulator is *ROSA-like*: its Denavit–Hartenberg parameters are plausible modeling values rather than measurements obtained from an official manufacturer CAD model.
- The surgical tool is represented as a rigid geometric element. Tissue deformation, tool bending, cutting forces, friction, and contact forces are not modeled.
- The controller is purely kinematic and does not include rigid-body dynamics, actuator dynamics, torque control, or force feedback.
- Jacobians are computed through finite differences rather than an analytic closed-form model, which simplifies implementation at the cost of additional computation and possible numerical sensitivity.
- The skull/corridor result validates the implemented geometric safety model only; it is not equivalent to validation against patient-specific anatomical data.
- The five runs repeat the same nominal scenario. Their strong agreement demonstrates repeatability but does not establish robustness over the robot workspace or under perturbations, noise, calibration errors, or anatomical variability.
- Tool dimensions and entry-target geometry remain simplified simulation parameters and should be matched to a representative neurosurgical instrument before claiming physical realism.

10 Conclusion

This project implements a custom task-priority double-RCM controller for a simulated ROSA-like six-degree-of-freedom robot. The controller supports four operating modes and a phased insertion sequence using damped least-squares inverse kinematics, null-space projection, directional joint-limit handling, and numerical integration substeps.

Across five nominal simulation runs, all nine declared checks pass in every run. The conservative settled results include 0.122 mm for Task 1 entry-RCM stability, 0.560 mm for Task 1 target tracking, 6.018 mm for Task 2 entry-cone tracking, 0.008 mm for Task 2 tip stabilization, 0.239 mm for Task 3 insertion RCM, 1.561 mm for the Task 3 final target, 0.283 mm for Task 4 entry-RCM stability, and 2.763 mm for Task 4 tip-cone tracking. No skull/corridor violation is recorded. The small between-run standard deviations show that the implemented controller is numerically repeatable under the tested nominal conditions.

These findings validate the intended kinematic behavior of the simulation, but they do not constitute clinical validation of the ROSA platform. Clinical or hardware-level conclusions would require an official robot model, realistic instrument and anatomy geometry, calibrated sensing, dynamic and contact effects, and a substantially broader experimental campaign.

Future Work

1. Replace the finite-difference Jacobians with analytic expressions and compare their numerical conditioning and computational cost.
2. Match the robot and tool geometry to verified ROSA and neurosurgical instrument parameters.
3. Introduce patient-specific anatomical models, collision geometry, tissue interaction, and force-aware safety constraints.
4. Perform randomized multi-run experiments with perturbed initial joint configurations, entry and target points, noise, calibration errors, and task durations.
5. Extend the directional active-set strategy from joint 4 to a general multi-joint constrained inverse-kinematics solver.
6. Validate the controller through hardware-in-the-loop experiments and, where possible, on an instrumented robotic platform.

References

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